

S01 System boundary and sheet map

Central Board Rev.1 | 2026-09-26 V2 harness and integrated coil driver | manual KiCad transcription reference



- S02A 5V input, eFuse and USB power sharing
- S02B six 5V branches and Teensy socket pin map
- S02M Teensy mechanical placement and pin orientation
- S03A/B CAN common circuit and connector mapping
- S04A/B E-stop 1ch + coil driver; page16 GUI / tactile ARM and separate RUN
- S05 external I/O general circuit and connector map
- S11 unresolved decisions and transcription checks

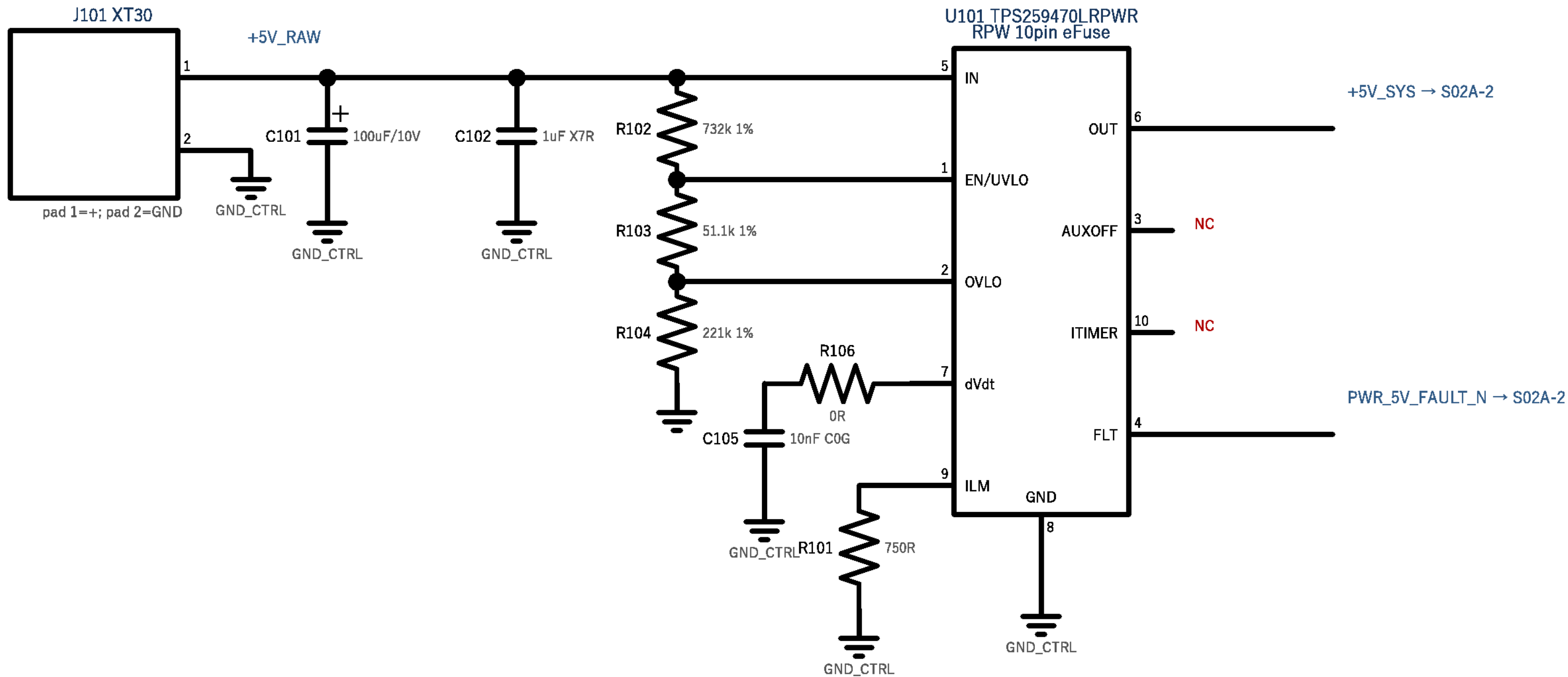
Priority: ADR + KiCad entry reference (2026-09-26 USB eFuse / GPIO VCC additions, pages 14-15). Existing 2026-09-17 PDF and legacy J402 driver text are superseded.

Dedicated CAN ports are horizontal JST GH3 (1=A, 2=B, 3=GND). Generic J323 remains 4pin.

No dedicated E228 main-contact feedback. Coil clamp/fuse and 24V return topology remain open on S11.

S02A-1 5V input and TPS259470 eFuse

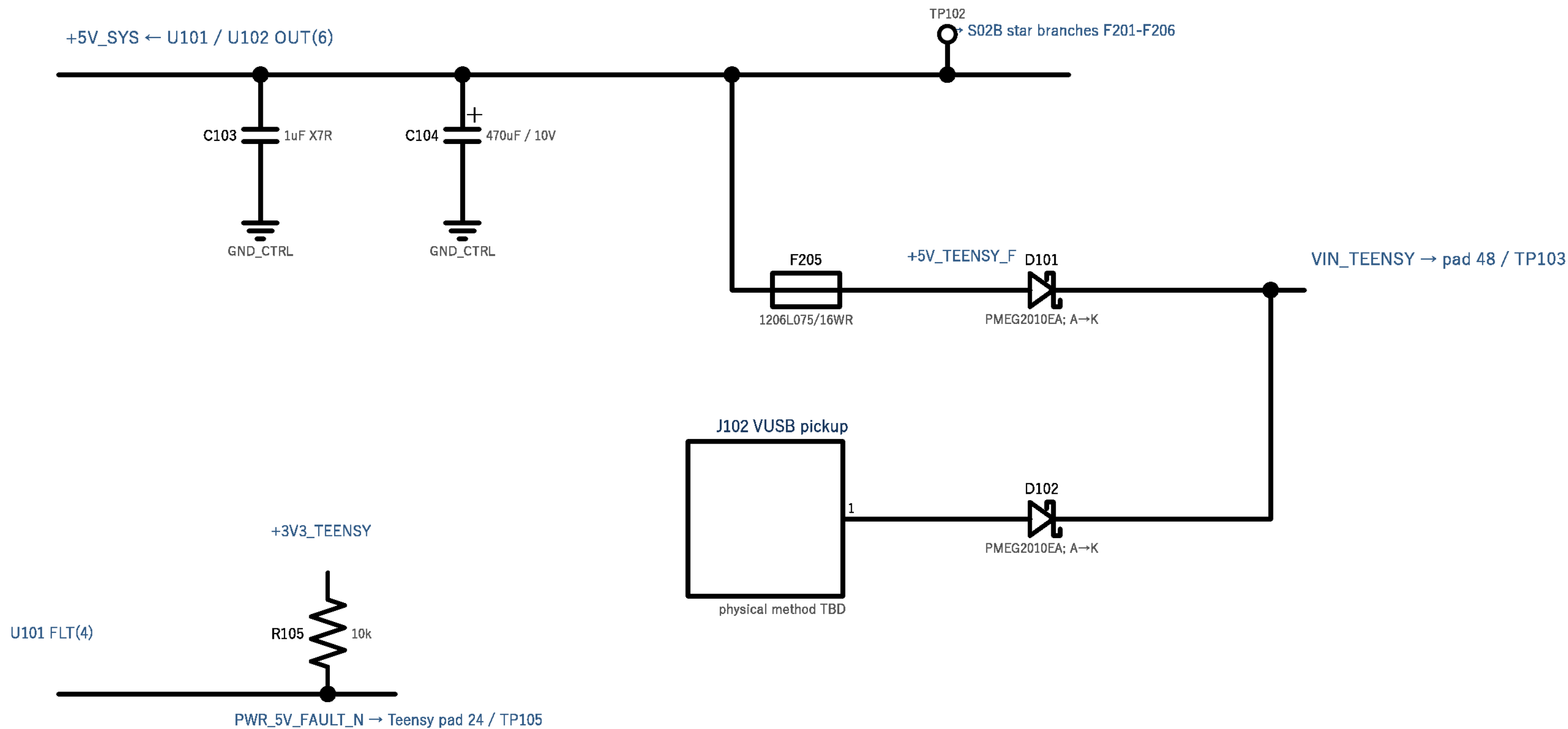
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Source: external SD-25B-5 → J101 XT30 → U101 TPS259470 → +5V_SYS. There is no 24V-to-5V buck on this PCB.
R102/R103/R104 form the UVLO/OVLO divider. U101 3=AUXOFF, 10=ITIMER are NC; 4=FLT has a 3.3V pull-up.

S02A-2 5V output and USB diode-OR to Teensy VIN

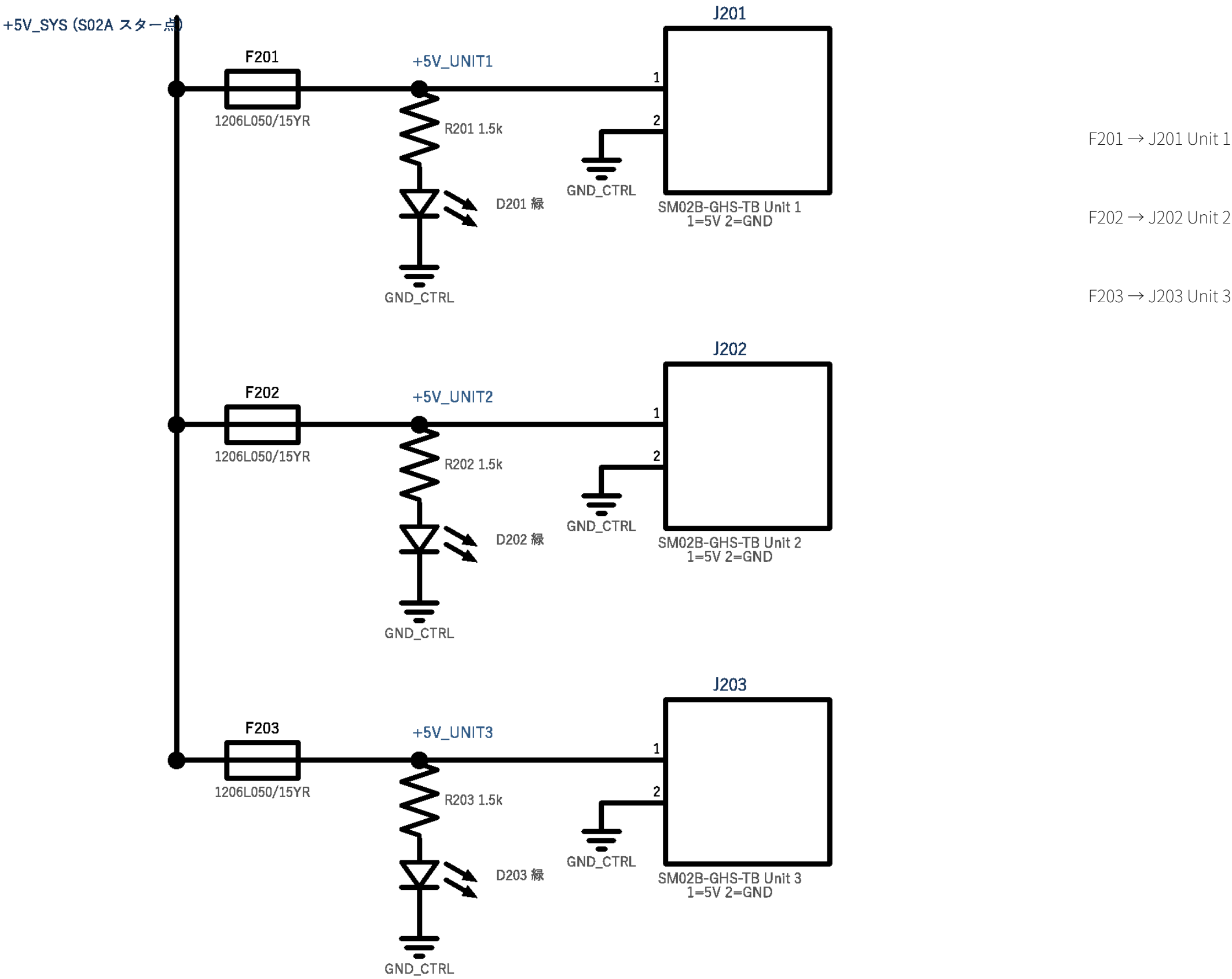
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D101 anode=F205 output; D102 anode=Teensy VUSB. Their cathodes join at VIN_TEENSY / pad 48.
Cut VUSB-VIN. J102 also feeds U102 IN(5) on S02A-USB (page 14); USB now supplies +5V_SYS and one node.

S02B-1 5V star branches to the three drive units

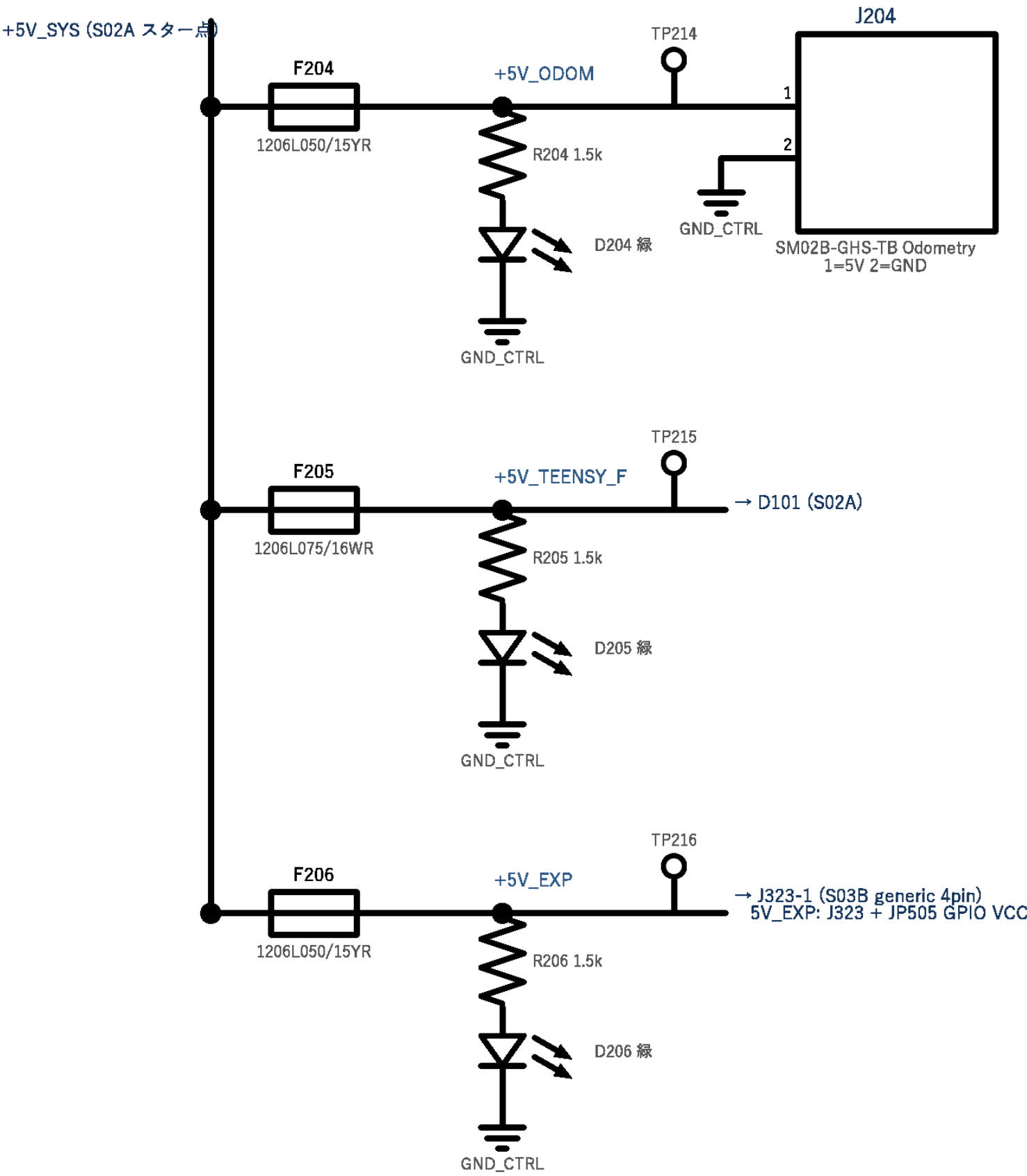
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J201-J203: pin 1=each protected 5V branch, pin 2=GND_CTRL. CAN3 GH3 carries its own GND reference on pin 3.
Unit TP211-TP213 omitted; measure at J201-J203 when needed. LEDs indicate power presence; keep protected nets separate.

S02B-2 5V branches to ODOM, Teensy and generic expansion

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F204 → J204 ODOM

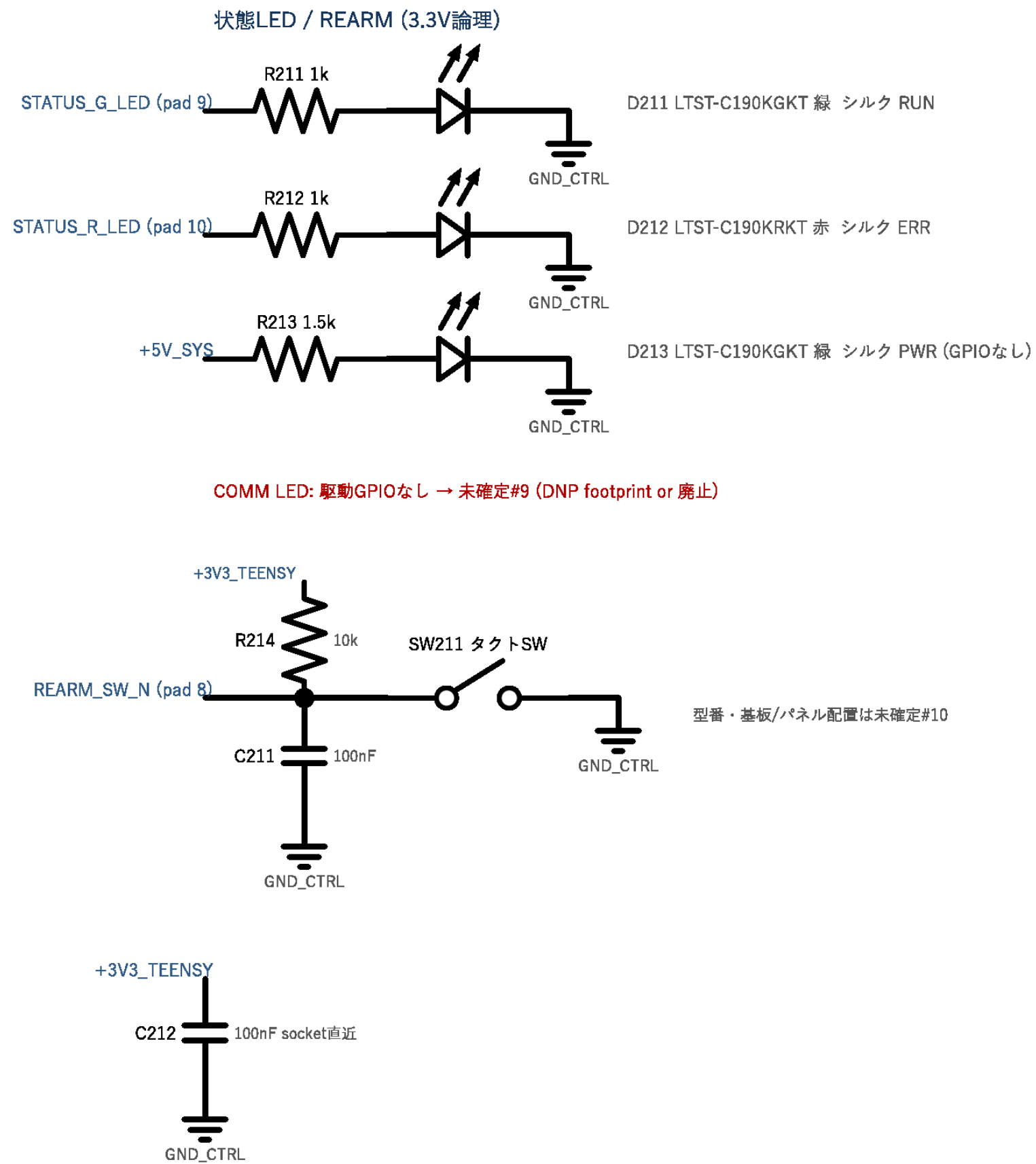
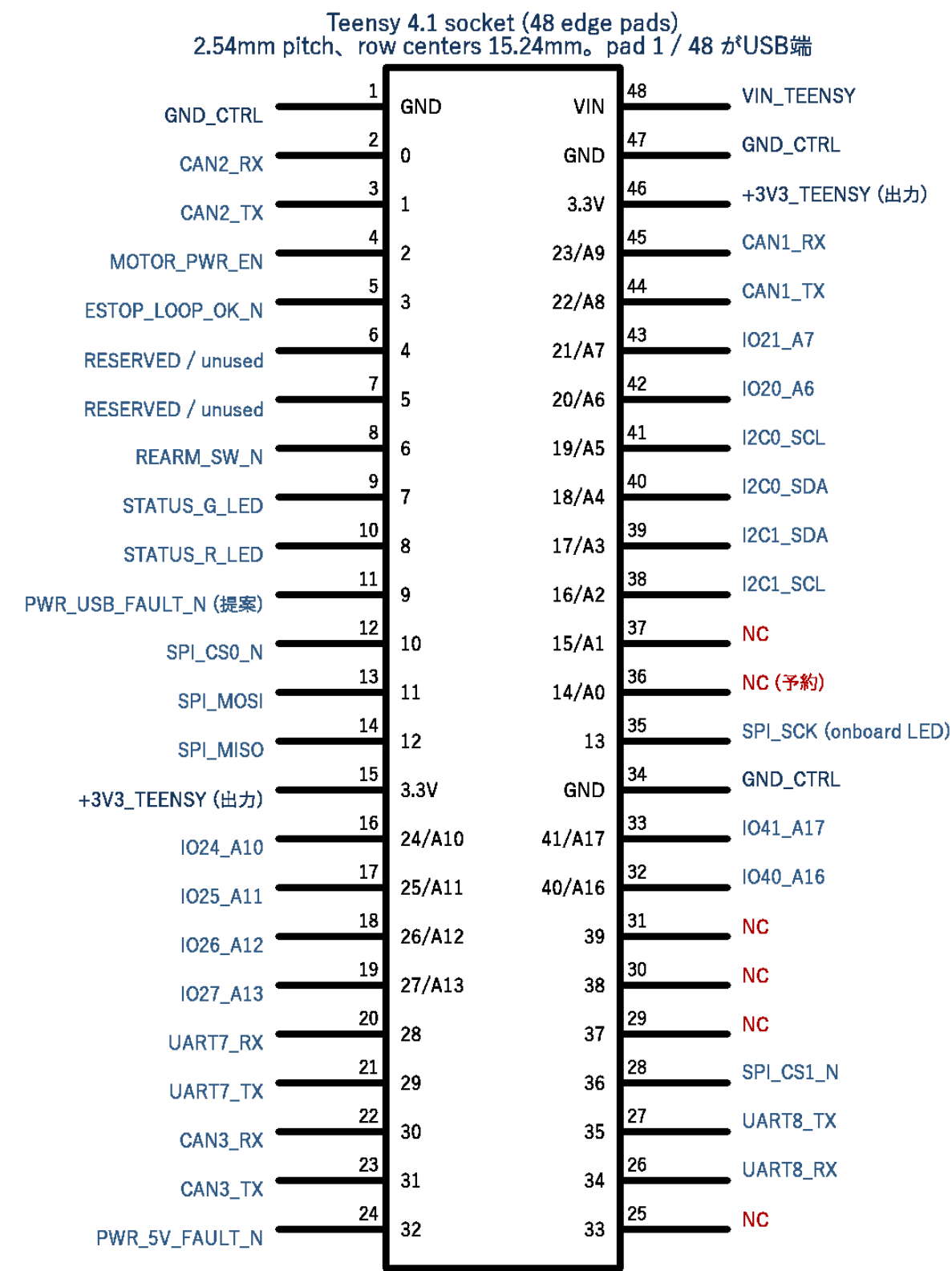
F205 → D101 Teensy VIN

F206 → J323 generic CAN2

J204 pin 2 carries ODOM supply GND. CAN1 GH3 also carries its GND reference on pin 3; F206 feeds generic J323 pin 1.
F205/F206 hold ratings need measured current/inrush review. Do not power self-powered LED/switching nodes through J323 by assumption.

S02C Teensy 4.1 socket - all 48 edge pads and local indicators

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MOTOR_PWR_EN is pad 4 / Teensy pin 2 → U402 A(2). Pad 11 / pin 9 = PWR_USB_FAULT_N (proposal); pad 36 / pin 14 remains NC.
SW211 stays tactile: REARM_SW_N pad8/pin6. GUI ARM over USB uses the SAME Teensy safety decision. See page16.

S02M Teensy socket placement - USB-end orientation

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PJRC mechanical drawing: 24 positions per row at 2.54 mm pitch; row center spacing is 15.24 mm.

17.78 mm is the Teensy board WIDTH, not the socket row spacing. Legacy central docs used it incorrectly.

Near/bottom row from USB end: Teensy pads 1 → 24 in order.

Far/top row from USB end: Teensy pads 48 → 25 in order.

USB end: pad 1 is left, pad 48 is right when viewed from component/top side.

Teensy socket footprint

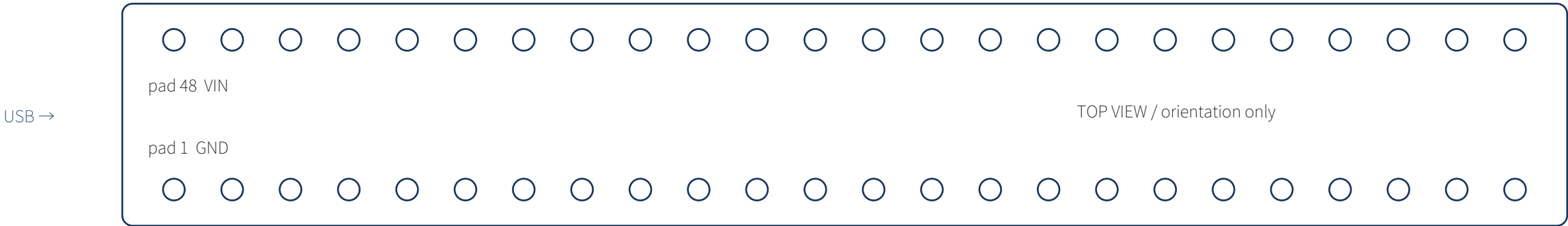
Prefer DifferentialSwerve:Teensy41_Socket_2x24
48-pad combined footprint; matching 48-pin symbol required
PJRC suggests Sullins PPPC241LFBN-RC / PPTC241LFBN-RC
Verify socket lead fit; USB / Program / microSD access

Existing PCB placement warning

Current central-board.kicad_pcb J3/J4 are about 33.3 mm apart and offset along the rows. They cannot mount a Teensy 4.1.
Do not copy that placement into the new central layout.
Keep components and exposed pads clear below the module.

Before fab: compare KiCad footprint dimensions with PJRC official drawing, then print at 1:1 scale.

Check pad 4 → MOTOR_PWR_EN and both 3.3V/GND/VIN pads against the socket pin map on S02C.



Mechanical source: https://www.pjrc.com/teensy/dimensions_teensy41.png ; PJRC socket: https://www.pjrc.com/store/socket_24x1.html

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Dedicated connectors are horizontal JST GH3: 1=COMM_A, 2=COMM_B, 3=GND. Generic J323 remains four-pin; see S03B.

S03B CAN connectors and GND reference paths

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Bus / nodes	Central port	Pin 1	Pin 2	Pin 3 / reference
CAN1 ODOM V2	J311/J312 GH3	SENSOR_COMM_A	SENSOR_COMM_B	GND_CTRL (pin 3)
CAN2 LED / switching	J321/J322 GH3	EXP_COMM_A	EXP_COMM_B	GND_CTRL (pin 3)
CAN3 Drive V2 x3	J331/J332 GH3	DRIVE_COMM_A	DRIVE_COMM_B	GND_CTRL (pin 3)
CAN2 generic	J323 4pin	+5V_EXP	GND_CTRL	3=A / 4=B

CAN1: Teensy TX pin 22 / pad 44 → U311 TXD; RX pin 23 / pad 45 ← U311 RXD.

CAN2: Teensy TX pin 1 / pad 3 → U321 TXD; RX pin 0 / pad 2 ← U321 RXD.

CAN3: Teensy TX pin 31 / pad 23 → U331 TXD; RX pin 30 / pad 22 ← U331 RXD.

All ports use COMM_A=CANH and COMM_B=CANL. J323 stays a distinct four-pin powered generic port.

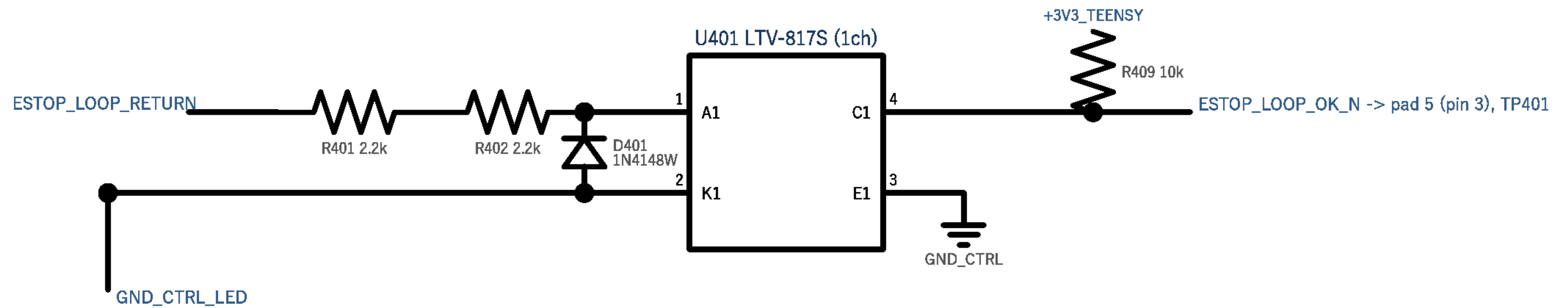
All dedicated CAN connectors: horizontal JST SM03B-GHS-TB; pin 1=A, 2=B, 3=GND. 5V/GND power wiring stays separate.

CAN reference GND must not carry LED/motor load current. Keep the three CAN systems separate.

Verify horizontal GH3 footprint pin-1 orientation, mating housing/contact and cable view. GH2 supply remains a separate connector.
J323 is a distinct powered four-pin generic CAN port; do not copy its pin order onto dedicated GH3 ports.

S04A E-stop NC chain and ONE loop-monitor channel

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J401 ESTOP_CTRL 4pin: 1=LOOP_FEED, 2=LOOP_RETURN, 3=+24V_CTRL_LED, 4=GND_CTRL_LED. Connector MPN TBD.

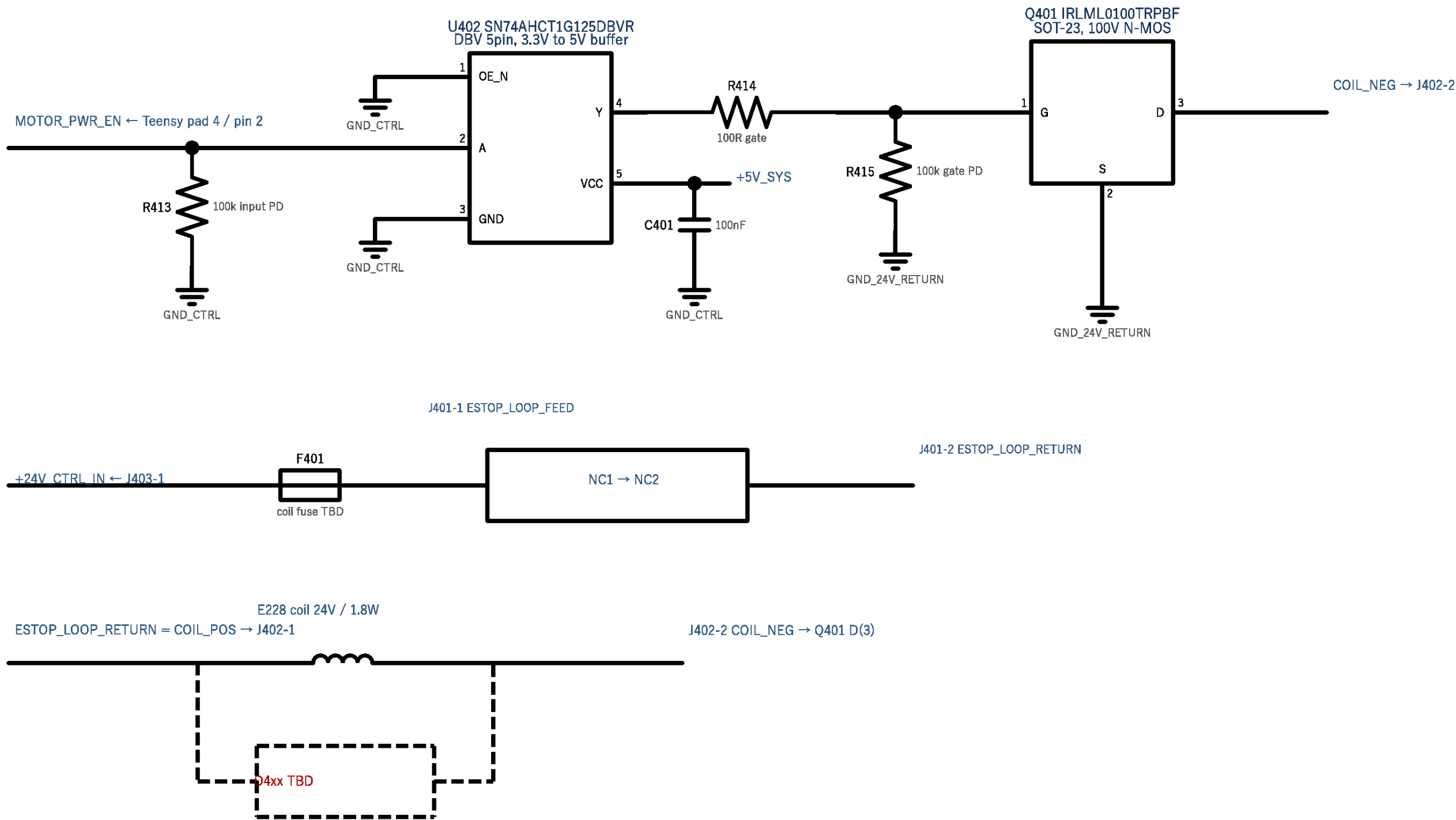
J403-1 → fuse TBD → J401-1 → NC1 → NC2 → J401-2 → COIL_POS/J402-1. This is a physical coil path independent of firmware.

U401 LTV-817S: LED A1/K2; transistor C4/E3. R401/R402=2.2k each; R409=10k to 3.3V.

D401 1N4148W is reverse parallel to the opto LED (anode at K). No individual button or spare monitoring. J402 coil = 2 wires.

S04B Central PCB integrated E228 coil driver

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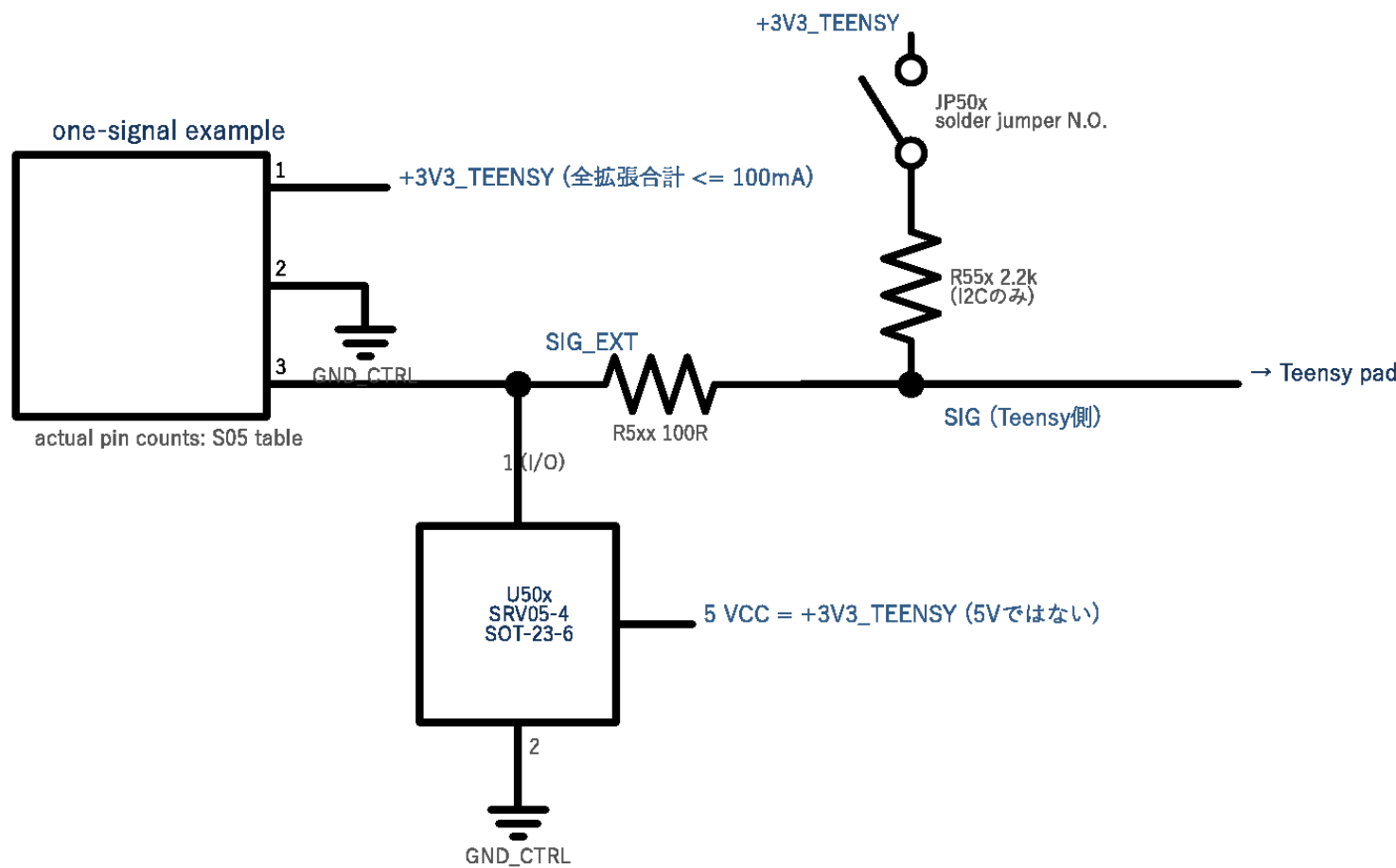


Dashed D4xx is NOT WIRE. STOP: clamp topology/value/polarity and release time TBD

Candidate U402 DBV pins: 1=OE_N tied low; 2=A from Teensy; 3=GND; 4=Y; 5=+5V_SYS. R413/R415=100k pull-downs; R414=100R.
Q401 IRLML0100 pin 1=G, 2=S, 3=D. J403-2=GND_24V_RETURN; J402-2=COIL_NEG. NC loop cuts COIL_POS even if Q401 is on.
F401 value/location, flyback clamp type/orientation, release time, J402/J403 MPN and GND bond are STOP/TBD before KiCad release.
GUI ARM or new SW211 press -> Teensy safety gate -> MOTOR_PWR_EN. E-stop release alone never re-arms. See page16.

S05 Protected external I/O common circuit

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Ref	Function / connector order
J501	Matek I2C 4pin: GND / SDA / SCL / +5V_SYS
J502/J503	I2C0 / I2C1: 3V3 / GND / SDA / SCL
J504/J505	UART7 / UART8: GND / TX / RX
J506	SPI 7pin: 3V3 / GND / SCK / MOSI / MISO / CS0_N / CS1_N
J507	GPIO/ADC 10pin: VCC_GPIO_EXT / GND / IO20/21/24/25/26/27/40/41

SRV05-4 x6: pins 1/3/4/6=protected lines, pin 2=GND, pin 5=+3V3_TEENSY.

Every external signal: connector → TVS → 100R series → Teensy. I2C pull-ups are jumper-selectable.

J507 VCC is selected by JP505 (page 15). Signals and TVS pin5 stay 3.3V; I2C/SPI VCC stays 3.3V.

Connector form factor and Matek +5V branch protection remain as earlier open items. Total external 3.3V load <=100mA.

S11 Open design decisions and manual KiCad checks

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CHECK 1 Dedicated CAN horizontal JST GH3 pin 1=A, 2=B, 3=GND; verify footprint, mate and cable orientation.

STOP 2 Coil fuse F401 rating/location and LED branch protection; confirm 24V supply maximum.

STOP 3 Coil flyback/clamp topology and part; measure E-stop release time and Q401 VDS peak.

STOP 4 J402 coil/J403 24V pin order, MPN and keying; verify E228 coil polarity if clamp is polarized.

STOP 5 5V GND, 24V coil return and CAN reference GND bond; ensure coil return does not cross quiet GND.

STOP 6 Teensy socket drill/pad/mechanical clearance and VUSB pad pickup method.

STOP 7 J501 Matek pull-up voltage; F205/F206 current and inrush; U401 package/land pattern.

STOP 8 Confirm U402 OE_N tie and power-up/down behavior on hardware; Reset/Hi-Z must keep Q401 off.

Transcription and review sequence

1 Reconcile every connector pin and every IC pin with this PDF, manufacturer pin diagrams and the final harness drawing.

2 Check socket orientation and 15.24 mm row spacing by 1:1 print; ignore current PCB J3/J4 placement.

3 Verify physical NC continuity from J403 via both E-stops to coil +; open either NC and measure coil de-energize.

4 Run KiCad ERC, then compare schematic netlist pin-by-pin to PCB pads; inspect all power pads and NC markers.

5 Run DRC and perform powered bench test: USB only, 5V only, combined, E-stop, reset, CAN GND under load.

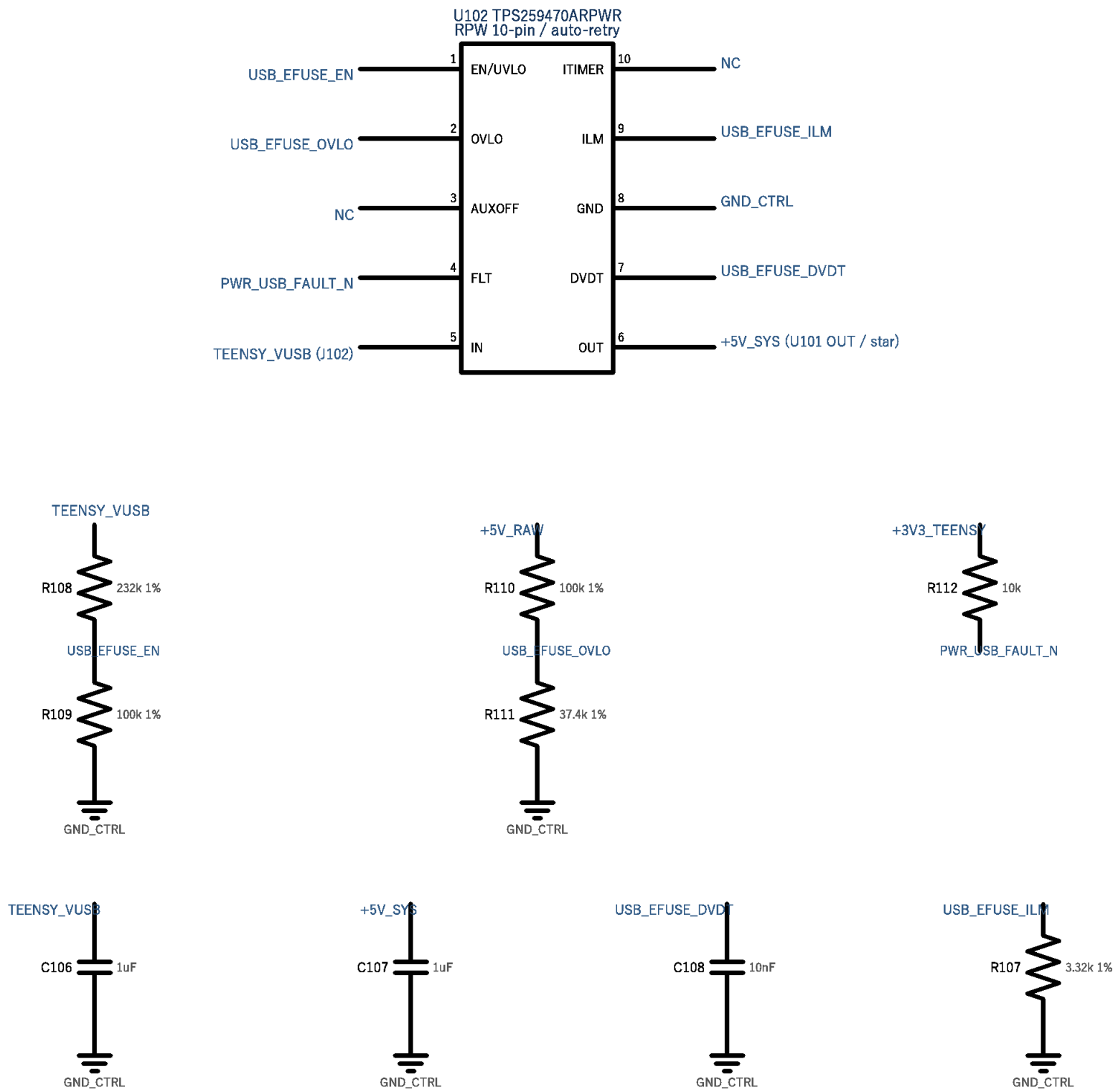
No main-contact monitor: when OFF command is issued and C620 replies cease, display OFF指令中・C620応答なし. CAN silence alone must never mean 遮断確認済み.

If central↔drive CAN itself is unavailable, show unknown/stale state, distinct from confirmed C620 nonresponse reported by drive MCU.

Legacy 2026-09-17 PDF has obsolete J402 external-driver text. Keep hard NC coil cut, reset/Hi-Z OFF and explicit re-arm.

S02A-USB USB eFuse - CAN and one 5V node without external power

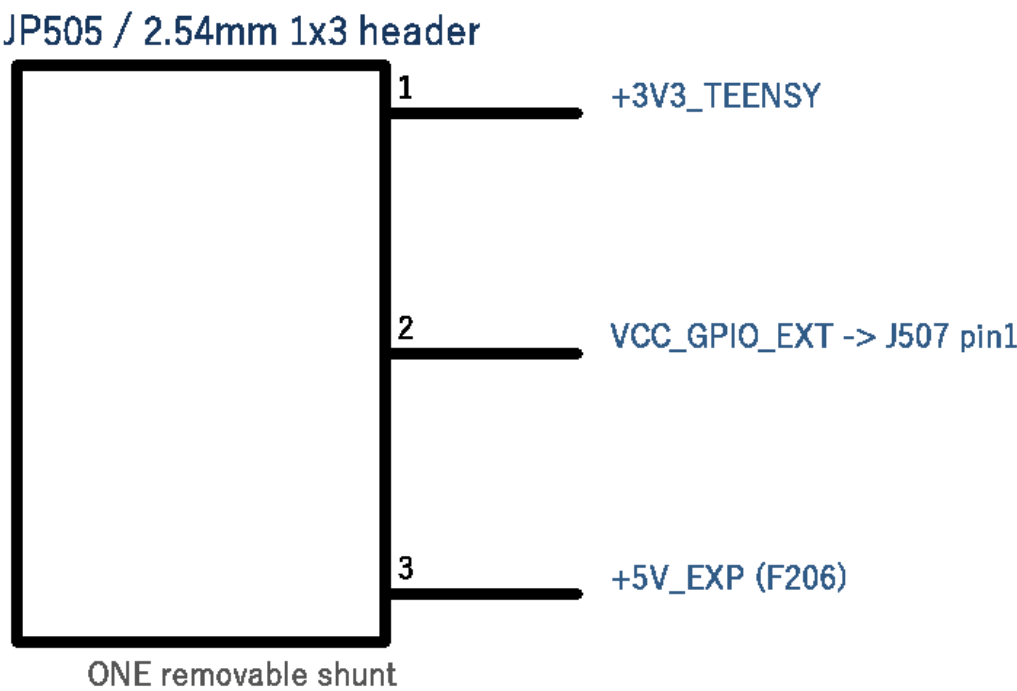
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U102 IN5=J102 VUSB; OUT6=+5V_SYS. U101 / U102 true RCB blocks reverse current; D102 direct VIN path remains.
TP106=PWR_USB_FAULT_N -> Teensy pad11 / GPIO9 (proposed). R112 pull-up goes to +3V3_TEENSY.
R107 3.32k: ILM 0.85 / 1.007 / 1.15A. R110/R111: external 5V >=4.41V switches U102 OFF (nominal).
U102 RPW footprint, VUSB pickup/current capacity and loaded +5V_SYS >=4.5V remain to be checked.
TI TPS25947 SLVSFC9C Rev.C; entry reference 3.1. USB3 / powered hub recommended; measure actual load.

S05-VCC GPIO supply selector - one header for all GPIO VCC pins

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1-2 = 3.3V (default)
2-3 = 5V
Only supply pin changes. IO stays 3.3V.

JP505 pin1=+3V3_TEENSY; pin2=VCC_GPIO_EXT; pin3=+5V_EXP. J507 pin1 connects to JP505 pin2.

Use one shunt: 1-2 = 3.3V default; 2-3 = 5V. Header/shunt purchase MPN remains unspecified.

Teensy IO remains 3.3V ONLY. U505/U506 SRV05-4 pin5 stays +3V3_TEENSY; ADC function remains.

I2C J502/J503 and SPI J506 supply pins remain fixed at 3.3V. Total external 3.3V load <=100mA.

5V GPIO supply shares F206 with J323; review combined current. Silk: VCC 3V3/5V and IO 3.3V ONLY.

ARM/RUN GUI and tactile button - shared Teensy re-arm decision

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ARM requests	Teensy shared safety gate	ARM accepted: power only
Robot display GUI -> mini PC -> USB-CDC OR SW211 tactile button -> pin6 / pad8 Both request ARM; neither drives the coil directly SW211 is retained for local maintenance	E-stop loop closed; no other stop/fault GUI + 3 drive MCUs + ODOM links fresh Motion command zero; NEW deliberate ARM action Reject old session / stop generation / sequence	MOTOR_PWR_EN -> U402 -> Q401 -> E228 coil J402 remains COIL_POS / COIL_NEG: 2 wires Keep motion outputs ZERO after ARM Require a separate NEW RUN request

E-stop open / link timeout / stop / fault -> DISARM, coil OFF, motion ZERO, invalidate prior ARM/RUN requests.

Releasing E-stop does NOT restore ARM. GUI click or new button press is required AFTER conditions become valid.

A held button cannot re-arm automatically: observe release while ready, then a new debounced press.

GUI reconnect / MCU restart requires a new boot_session and fresh ARM. GUI displays only fresh Teensy state.

ARM never starts motion. RUN must be newer than ARM; check C620 startup/fresh responses and normal run prerequisites.

OFF-state C620 silence is expected: it is NOT an ARM prerequisite. Drive-MCU heartbeat and C620 response are distinct.

No contactor auxiliary-contact input or MOTOR_PWR_SENSE. CAN silence does not prove main-contact opening.

NC1 + NC2 physically interrupt coil power without MCU/CAN. Software ARM cannot override the NC chain.

Implementation: hardware-independent ArmController added. Teensy GPIO/USB adapter and robot-display GUI are pending.

No build / bench validation yet. Integration must implement timeouts, debounce, session IDs and C620 startup checks.

Source: docs/software/MINIPC_GUI_AND_TEENSY.md; central_firmware/include/control/arm_controller.h (2026-09-26).